

Solution Architecture

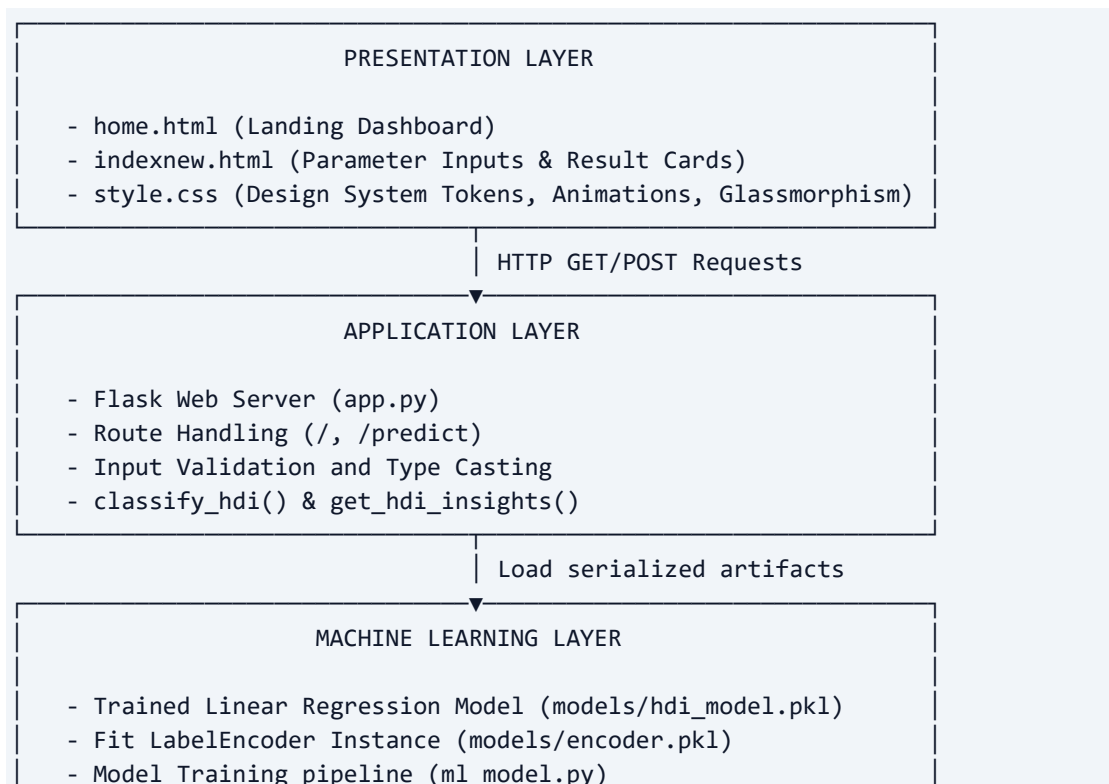
DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor
MAXIMUM MARKS	4 MARKS

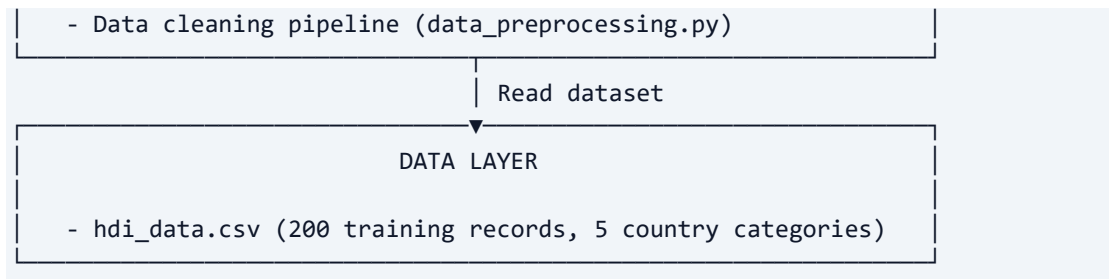
1. Overview

This document specifies the software architecture design for the HDI Prediction System, describing the system layers, directory structure, and component interactions.

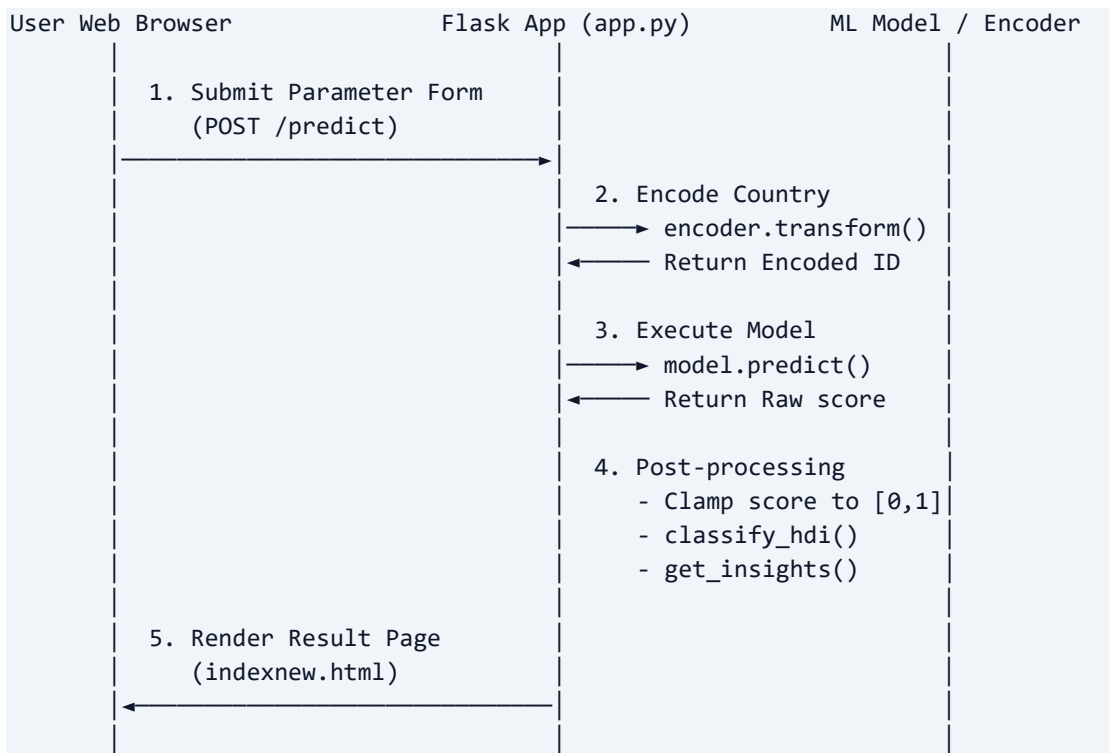
2. High-Level System Layers

The system follows a multi-tier architecture to separate presentation, business logic, model execution, and data storage.



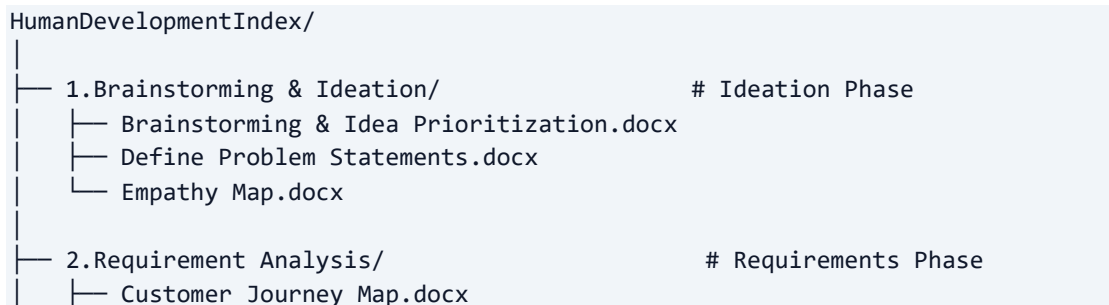


3. Detailed Component Interaction



4. Directory Structure Mapping

To ensure a clean layout, the project structure is organized as follows:



└─ Data Flow Diagram.docx └─ Solution Requirements.docx └─ Technology Stack.docx	
└─ 3.Project Design Phase/ └─ Problem - Solution Fit.docx └─ Proposed Solution.docx └─ Solution Architecture.docx	# Architectural Design
└─ 4.Project Planning Phase/ └─ Project Planning.docx	# Project Planning
└─ 5.Project Development Phase/ └─ Code-Layout, Readability and Reusability.docx └─ Coding & Solution.docx └─ No. of Functional Features Included in the Solution.docx └─ code/ └─ app.py └─ ml_model.py └─ data_preprocessing.py └─ visualization.py └─ eda_visualizations.py └─ feature_selection.py └─ requirements.txt └─ schema.sql └─ test_app.py └─ data/ └─ hdi_data.csv └─ models/ └─ hdi_model.pkl └─ encoder.pkl └─ static/ └─ style.css └─ templates/ └─ home.html └─ index.html └─ indexnew.html	# Project Code & Docs # Active Codebase
└─ 6.Project Testing/ └─ Acceptance Testing.docx └─ Functional & Performance Testing.docx	# Testing & Quality Assurance
└─ 7.Project Documentation/ └─ Project Documentation.docx └─ Project Executable Files.docx	# Final Reports & Manuals
└─ 8.Project Demonstration/ └─ Communication.docx └─ Demonstration of Proposed Features.docx	# Demonstration Phase

- └─ Project Demo Planning.docx
- └─ README.docx
- └─ Scalability & Future Plan.docx
- └─ Team Involvement in Demonstration.docx